

RobotSTAR: Next-Scale Autoregressive Sign Language Motion Translation for Humanoid Robots

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Abstract—Providing sign language interpretation for public broadcasts and press conferences is labor-intensive and difficult to scale, motivating robotic signing as a complementary means of improving accessibility for deaf and hard-of-hearing communities. We present RobotSTAR, a sign language production framework for embodied signing systems that supports both written and speech inputs. Conventional autoregressive approaches generate sign motion as a flat token sequence at a single temporal resolution, making it difficult to simultaneously model global body motion and fine-grained hand articulation. RobotSTAR instead introduces next-scale autoregression for sign language generation, producing motion progressively from coarse to fine temporal resolutions. Coarse scales establish the global temporal and motion structure, while finer scales progressively refine local dynamics, hand articulation, and body–hand coordination. The resulting representation is decoded into continuous human-centric motion that can serve as input to downstream embodiment-specific retargeting. Experiments demonstrate state-of-the-art geometric accuracy and strong semantic and motion-dynamics performance, highlighting the effectiveness of coarse-to-fine autoregression for embodied sign language generation.

Index Terms—sign language production, embodied motion generation, human–robot interaction

I. INTRODUCTION

Sign language interpretation is essential for making televised news, press conferences, emergency briefings, and other public information accessible to deaf and hard-of-hearing communities. These settings typically depend on professional interpreters, which can make continuous and large-scale coverage difficult to sustain. Robotic signing offers a complementary direction: a spoken or written message can be converted into sign motion and subsequently transferred to an embodied platform for repeated or on-demand information delivery. A key challenge in such a system is therefore to generate accurate, temporally coherent sign motion from language.

Sign languages are natural languages whose realization depends on tightly coordinated handshape, arm motion, body posture, and non-manual signals. Existing sign language production (SLP) methods generate continuous poses through regression [1], [2], model motion with diffusion [3], or discretize motion for language-model-based generation [4]–[7]. Discrete approaches provide a convenient interface to pretrained language models, but most operate on a flat token sequence at a single temporal resolution. This forces a single token stream to capture heterogeneous motion patterns across the body and hands, ranging from large-scale arm and torso movements to fine-grained finger articulation. This issue is

important for robotic signing, which requires accurate and coordinated generation of both body motion and fine-grained hand articulation.

We address this problem with **RobotSTAR**, a coarse-to-fine autoregressive framework for robot-oriented SLP. We separately tokenize the upper body and bilateral hands using finite scalar quantization (FSQ) [8], while preserving temporal synchronization across the three streams. Instead of predicting the full-resolution motion directly, RobotSTAR generates a hierarchy of temporal scales following the next-scale autoregressive principle [9], [10]. Coarse scales establish the overall temporal and motion structure, while finer scales progressively recover local dynamics and detailed hand articulation. To improve robustness to errors propagated across scales, we introduce self-conditioning (SC) and generic context corruption (GCC) during training.

Our contributions are:

- We introduce a **part-wise FSQ motion representation** that models the distinct statistics of upper-body and hand motion while preserving their temporal coordination.
- We formulate sign language production as **next-scale autoregression**, a coarse-to-fine hierarchy that progressively refines motion detail.
- We demonstrate **state-of-the-art geometric accuracy**, strong semantic and motion-dynamics performance, improved ratings from fluent signers, and physical execution on a humanoid robot.

II. RELATED WORK

Sign language production. Early SLP methods generate continuous pose sequences from text or gloss [1], [2], while later work extends generation to holistic 3D signing motion [3], [11], [12]. Recent methods increasingly discretize motion for language-model-based generation: MotionGPT [4] establishes this paradigm for general human motion, while T2S-GPT [5], SOKE [6], and M³T [7] adapt it to sign generation with increasingly structured body, hand, and facial representations. In parallel, direct sign-video generation has progressed from pose-conditioned synthesis [13], [14] to end-to-end video generation [15], [16]. These methods emphasize visual realism, whereas robotic embodiment requires explicit 3D motion that can be retargeted and physically executed. RobotSTAR addresses this setting through part-wise FSQ and coarse-to-fine motion generation.

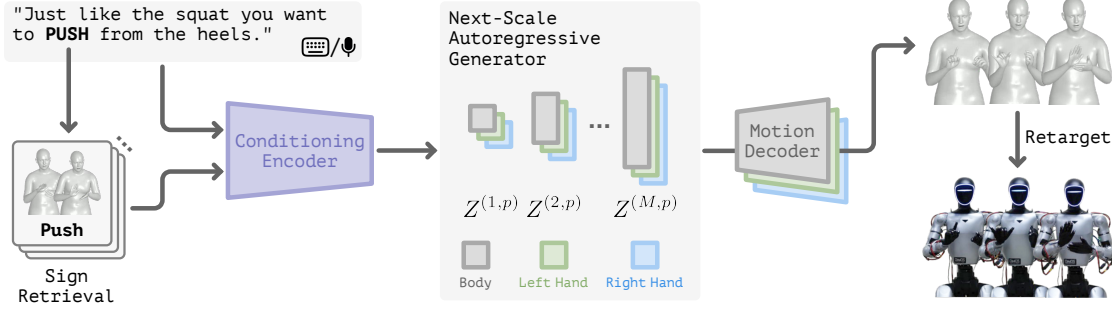


Fig. 1. **Overview of RobotSTAR.** Speech is transcribed to text, while written input is used directly. Language features and retrieved lexical cues condition synchronized body and hand FSQ tokens generated from coarse to fine and decoded into continuous human-centric motion, which is subsequently retargeted to a humanoid robot for physical execution.

Coarse-to-fine autoregression. Next-scale autoregression models global structure before progressively introducing finer detail, as explored in image and video generation [9], [17], [18], while MoScale [10] applies next-scale prediction to text-conditioned human motion. RobotSTAR extends this paradigm to synchronized body and bilateral-hand streams for sign generation.

Embodied robotic signing. Prior robotic signing systems mainly focus on transferring or interpreting sign motion. Human-to-robot ASL transfer [19] retargets captured demonstrations to a bimanual robot, SignVLA [20] maps finger-spelled signs to manipulation commands, and SignBot [21] builds on SOKE [6] as its sign-generation backbone to study bidirectional sign language communication between humans and humanoid robots. In contrast, we formulate speech/text-to-robotic signing as an embodied translation task, converting spoken or written language into coordinated sign motion for humanoid robotic embodiment.

III. METHOD

A. Part-Wise Multi-Scale Motion Tokens

A normalized sign sequence is represented as

$$S = [S^B, S^{LH}, S^{RH}] \in \mathbb{R}^{T \times d}, \quad (1)$$

where S^B contains upper-body and facial-expression parameters, while S^{LH} and S^{RH} represent the left and right hands. We use dedicated temporal encoders and decoders because body motion and fine-grained hand articulation exhibit different statistics. For part $p \in \mathcal{P} = \{B, LH, RH\}$, an encoder $\mathcal{E}_p$ produces a latent sequence. Each latent vector is projected into a low-dimensional FSQ space and independently quantized along each scalar dimension,

$$\hat{u}_{\tau,j}^p = Q_{n_{p,j}}(u_{\tau,j}^p), \quad K_p = \prod_{j=1}^{d_q} n_{p,j}, \quad (2)$$

then mixed-radix encoding converts the quantized tuple into one token from K_p valid states. At the finest scale, unlike VQ-VAE [22], FSQ uses a fixed Cartesian grid rather than a learned nearest-neighbor codebook [8].

Only the finest token sequence $Z^{(M,p)}$ is decoded to reconstruct motion. Coarser targets $Z^{(m,p)}$, $m < M$, are

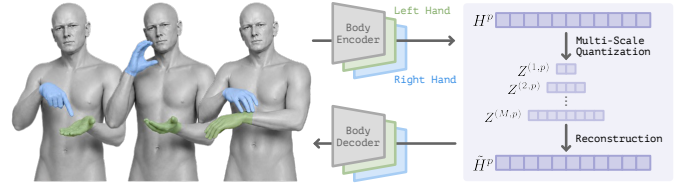


Fig. 2. **Part-wise FSQ pyramid.** Dedicated encoders tokenize the body and hands. Coarse targets are obtained by pooling finest-scale code vectors and projecting them back to the valid FSQ grid; only the finest scale is decoded for reconstruction.

constructed by dequantizing the finest tokens, temporally average-pooling their implicit code vectors to length L_m , and projecting the pooled vectors back to the valid FSQ grid. Every coarse scale is derived independently from the finest scale rather than recursively, yielding synchronized lengths $L_1 < \dots < L_M$ for all three parts. This construction provides coarse-scale supervision while avoiding recursive error accumulation across scales.

B. Next-Scale Autoregressive Generation

Let x be written text or a transcript from a fixed ASR front end. A pretrained multilingual language model [23] encodes x , and the resulting features are augmented with dictionary-retrieved lexical sign tokens following retrieval-enhanced SLP [6]; we denote the resulting condition by C_x . A length predictor estimates L_M , and fixed ratios determine the coarser lengths. Instead of flattening all motion tokens, RobotSTAR factorizes generation across scales and time:

$$p_{\theta}(Z | x) = \prod_{m=1}^M \prod_{\tau=1}^{L_m} \prod_{p \in \mathcal{P}} p_{\theta,p} \left(z_{\tau}^{(m,p)} \mid C_x, Z^{(<m)}, Z_{<\tau}^{(m)} \right). \quad (3)$$

Scales are generated sequentially from coarse to fine. Within each scale, temporal positions remain autoregressive, while part-specific heads predict synchronized body and bilateral-hand tokens in parallel from a shared hidden state. A hierarchical causal mask exposes the language condition, all coarser scales, and the current-scale prefix, while blocking future positions and finer scales. Rather than assigning a fixed linguistic role to each level, the hierarchy progressively increases temporal resolution to capture finer motion detail.

TABLE I
QUANTITATIVE COMPARISON ON THE COMMON HOW2SIGN TEST SUBSET. GT IS A REFERENCE AND EXCLUDED FROM RANKING. B- n DENOTES BACK-TRANSLATION BLEU- n . BEST AND SECOND-BEST GENERATION RESULTS ARE **BOLD** AND UNDERLINED.

Method	DTW-PA-JPE ↓		DTW-JPE ↓		B-T BLEU ↑				Motion dynamics		
	Body	Hand	Body	Hand	B-1	B-2	B-3	B-4	Dyn.↔ 0.8	Mag.↔ 1	Freeze↓
GT	—	—	—	—	22.45	8.59	3.95	1.96	0.804	1.000	0.036
PT	10.85	2.66	10.84	10.97	13.44	4.69	1.99	0.91	0.245	0.484	0.232
NAR	<u>4.21</u>	<u>1.79</u>	<u>4.14</u>	<u>5.70</u>	6.81	2.10	0.88	0.42	0.000	0.004	1.000
S-MotionGPT	11.84	3.92	11.91	13.30	12.01	4.33	1.99	1.03	0.005	0.016	0.294
T2S-GPT	4.35	1.82	4.27	5.78	7.13	2.79	1.28	0.60	0.000	0.000	1.000
SOKE	5.73	2.08	5.73	7.26	<u>17.25</u>	<u>6.31</u>	<u>2.86</u>	1.39	<u>0.388</u>	<u>0.515</u>	0.163
RobotSTAR	3.96	1.52	3.87	5.34	20.08	7.12	3.04	<u>1.36</u>	0.432	0.523	<u>0.187</u>

C. Robust Cross-Scale Conditioning and Output Interface

Teacher forcing supplies clean preceding scales during training, whereas inference supplies model predictions. SC periodically replaces clean preceding-scale targets with stop-gradient model predictions, exposing finer scales to structured errors produced by the generator itself. GCC independently perturbs a small subset of preceding-scale tokens, encouraging robustness to broader context noise. Both strategies modify only the training context and add no inference-time modules.

The generator is optimized with scale- and part-weighted token losses together with duration and text–motion consistency terms:

$$\mathcal{L} = \sum_{m=1}^M \lambda_m \sum_{p \in \mathcal{P}} \alpha_p \mathcal{L}_{CE}^{(m,p)} + \lambda_{\text{len}} \mathcal{L}_{\text{len}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}}. \quad (4)$$

After generation, the finest body and hand tokens are decoded into continuous trajectories. For embodiment, SMPL-X upper-body targets are mapped to the robot in a torso-centered frame under morphology, joint-limit, and temporal-continuity constraints, while MANO hand articulation is retargeted independently from palm-normalized geometry and tracked by a platform-specific controller. Unsupported facial channels are omitted during execution.

IV. EXPERIMENTS

A. Setup

Data and input. We evaluate on How2Sign [24] using the official training, validation, and test splits. For benchmark evaluation, paired English transcripts are used as input to isolate sign-motion generation from upstream speech-recognition errors. Upper-body motion is reconstructed with WHAC [25], and MANO-compatible hands [26] with WiLoR [27]. The resulting 133-D representation contains upper-body, bilateral-hand, and facial-expression parameters; camera, lower-body, and identity-dependent variables are discarded. For speech input, we use the frozen 2B-parameter Granite Speech 4.1 model [28] to transcribe audio before feeding the resulting text to RobotSTAR.

Generated motions are retargeted to a Unitree G1 equipped with dual 20-DoF Wuji hands for physical execution.

Metrics. Following prior 3D SLP work [6], [7], dynamic time warping (DTW) [29] aligns sequences of different lengths. We report joint-position error before and after Procrustes



Fig. 3. **Qualitative comparison on How2Sign.** Representative frames from ground truth, SOKE, and RobotSTAR.

alignment (DTW-JPE and DTW-PA-JPE) for body and hands. A frozen sign-to-text model provides auxiliary back-translation BLEU [30]; Dynamic Degree, Magnitude Ratio, and Freeze-tail characterize motion dynamics and degeneration. All methods are evaluated on the same test subset with identical topology, alignment, and metric implementations.

B. Main Results

Table I shows that RobotSTAR achieves the lowest error across all four geometric metrics. These results demonstrate the benefit of part-wise motion modeling and coarse-to-fine temporal generation, particularly for fine-grained hand articulation.

For semantic consistency, RobotSTAR obtains the best B-1, B-2, and B-3 scores and the second-best B-4. Its Dynamic Degree and Magnitude Ratio are also closest to the reference targets, while maintaining a low Freeze-tail value. Since back-translation depends on a separate recognition model, BLEU is treated as an auxiliary relative metric rather than a direct measure of sign correctness.

Qualitatively, Fig. 3 shows that RobotSTAR more closely follows the reference arm trajectories and produces clearer bilateral hand configurations than SOKE, particularly when handshape changes must remain coordinated with larger body and arm movements.

Real-robot execution. We further deploy RobotSTAR-generated motions on a Unitree G1 equipped with dual 20-DoF Wuji hands. After morphology-aware retargeting, the physical robot reproduces coordinated upper-body and dexterous-hand movements across representative sequences, demonstrating the feasibility of transferring the generated human-centric signs to a humanoid platform.

TABLE II
ABLATION OF SELF-CONDITIONING (SC) AND GENERIC CONTEXT CORRUPTION (GCC). BEST AND SECOND-BEST RESULTS ARE SHOWN IN BOLD AND UNDERLINED.

Configuration	Training		DTW-PA-JPE ↓		DTW-JPE ↓		B-T BLEU ↑				Motion Dynamics		
	SC	GCC	Body	Hand	Body	Hand	B-1	B-2	B-3	B-4	Dynamic ↔ 0.8	Magnitude ↔ 1	Freeze-tail ↓
Neither	×	×	4.19	1.85	4.06	5.90	8.85	3.02	1.37	0.63	0.13	0.23	0.16
GCC only	×	✓	4.25	1.96	4.10	6.04	8.17	2.74	1.23	0.55	0.15	0.25	0.23
SC only	✓	×	3.92	<u>1.54</u>	3.82	5.07	<u>17.93</u>	<u>6.34</u>	<u>2.67</u>	<u>1.14</u>	<u>0.34</u>	<u>0.47</u>	0.31
Ours (SC+GCC)	✓	✓	<u>3.96</u>	1.52	<u>3.87</u>	<u>5.34</u>	20.08	7.12	3.04	1.36	0.43	0.52	<u>0.19</u>

TABLE III
EFFECT OF NEXT-SCALE AUTOREGRESSION.

Variant	Scales	DTW-PA-JPE ↓		DTW-JPE ↓	
		Body	Hand	Body	Hand
Single-scale AR	{1}	4.66	2.56	4.43	6.81
Next-scale	{8, 4, 2, 1}	3.96	1.52	3.87	5.34

TABLE IV
FSQ VERSUS VQ-VAE AT MATCHED COMPRESSION.

Quantizer	Reconstruction		Generation			
	N-MAE	N-RMSE	PA-B	PA-H	JPE-B	JPE-H
FSQ	0.522	0.705	3.96	1.52	3.87	5.34
VQ-VAE	0.410	0.557	4.75	2.18	4.62	6.32

C. Ablation Studies

Next-scale generation. Table III compares the complete four-scale hierarchy with conventional full-resolution single-scale autoregression. Next-scale generation reduces DTW-PA-JPE by 14.9% for the body and 40.4% for the hands, and DTW-JPE by 12.7% and 21.5%, respectively. The substantially larger improvement on hand motion supports coarse-to-fine refinement for fine-grained articulation.

Motion tokenization. Table IV compares FSQ and VQ-VAE under the same part-wise architecture and a matched 740.17× compression ratio. VQ-VAE provides better oracle reconstruction, whereas FSQ consistently produces more accurate generated motion. This result reveals a reconstruction–predictability trade-off: better reconstruction does not necessarily yield better targets for autoregressive generation.

Cross-scale robustness. Table II studies SC and GCC. SC provides the main geometric improvement by exposing finer scales to model-generated histories. Adding GCC to SC introduces a modest geometric trade-off but consistently improves semantic consistency and motion dynamics. We therefore adopt SC+GCC as the final configuration.

D. User Study

We conduct a blinded user study with four fluent signers using 15 held-out sentences from How2Sign. For each sentence, participants view the ground-truth motion, SOKE output, and RobotSTAR output in randomized order with method identities hidden. Each motion receives a single overall score on a 1–10 scale, jointly considering semantic accuracy, hand-shape clarity, motion naturalness, and temporal fluency. Ratings are first averaged across the four participants for each sentence,

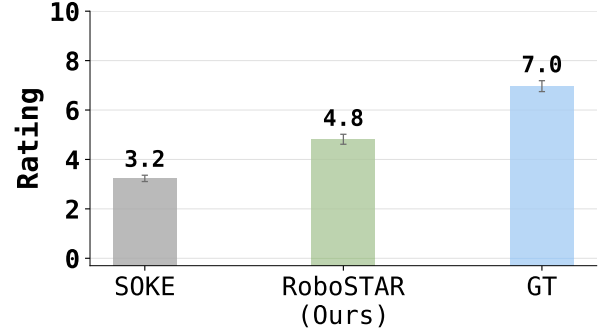


Fig. 4. **User study.** Mean ratings; error bars denote SEM.

and we report the mean and standard error of the mean (SEM) across the 15 sentence-level scores.

As shown in Fig. 4, RobotSTAR receives higher mean ratings than SOKE and narrows the perceptual gap to the ground-truth motions, providing complementary perceptual evidence beyond the automatic geometric and semantic metrics.

V. LIMITATIONS AND FUTURE WORK

Our evaluation remains limited to How2Sign/ASL and may inherit dataset- and signer-specific biases. Monocular 3D reconstruction may introduce errors, particularly for occluded or fine-grained hand poses, and the main benchmark uses ground-truth transcripts rather than evaluating ASR error propagation end to end. Finally, the target robot lacks facial and lip actuation, limiting non-manual cues. Future work will broaden sign-language coverage, improve 3D supervision, evaluate the complete speech-to-sign pipeline, and explore richer embodiments.

VI. CONCLUSION

In this paper, we presented **RobotSTAR**, a framework for speech/text-to-robotic sign language production. At its core, RobotSTAR organizes sign generation as a coarse-to-fine temporal pyramid, progressively refining global motion structure into detailed body and hand articulation. This hierarchical formulation achieves state-of-the-art performance on How2Sign, highlighting the benefit of next-scale generation for sign motion with complex body–hand coordination. Beyond benchmark performance, RobotSTAR advances speech/text-to-robotic signing as an embodied translation task, providing a foundation for accessible information delivery in public broadcasts, press conferences, and related settings.

REFERENCES

- [1] B. Saunders, N. C. Camgöz, and R. Bowden, “Progressive transformers for end-to-end sign language production,” in *European Conference on Computer Vision*, A. Vedaldi, H. Bischof, T. Brox, and J. Frahm, Eds., 2020.
- [2] E. J. Hwang, J. Kim, and J. C. Park, “Non-autoregressive sign language production with gaussian space,” in *British Machine Vision Conference*, 2021.
- [3] V. Baltatzis, R. A. Potamias, E. Ververas, G. Sun, J. Deng, and S. Zafeiriou, “Neural sign actors: A diffusion model for 3d sign language production from text,” in *Conference on Computer Vision and Pattern Recognition*, 2024.
- [4] B. Jiang, X. Chen, W. Liu, J. Yu, G. Yu, and T. Chen, “Motiongpt: Human motion as a foreign language,” in *Conference and Workshop on Neural Information Processing Systems*, 2023.
- [5] A. Yin, H. Li, K. Shen, S. Tang, and Y. Zhuang, “T2S-GPT: dynamic vector quantization for autoregressive sign language production from text,” in *Annual Meeting of the Association for Computational Linguistics*, 2024.
- [6] R. Zuo, R. A. Potamias, E. Ververas, J. Deng, and S. Zafeiriou, “Signs as tokens: A retrieval-enhanced multilingual sign language generator,” in *International Conference on Computer Vision*, 2025.
- [7] A. Symeonidis-Herzig, J. Low, O. M. Sincan, and R. Bowden, “M3T: discrete multi-modal motion tokens for sign language production,” *arXiv preprint*, 2026.
- [8] F. Mentzer, D. Minnen, E. Agustsson, and M. Tschanen, “Finite scalar quantization: VQ-VAE made simple,” in *International Conference on Learning Representations*, 2024.
- [9] K. Tian, Y. Jiang, Z. Yuan, B. Peng, and L. Wang, “Visual autoregressive modeling: Scalable image generation via next-scale prediction,” in *Conference and Workshop on Neural Information Processing Systems*, 2024.
- [10] Z. Zheng, S. Jin, L. Liu, and M. Zhao, “Next-scale autoregressive models for text-to-motion generation,” in *Conference on Computer Vision and Pattern Recognition*, 2026.
- [11] Z. Yu, S. Huang, Y. Cheng, and T. Birdal, “Signavatars: A large-scale 3d sign language holistic motion dataset and benchmark,” in *European Conference on Computer Vision*, 2024.
- [12] L. Dong, X. Wang, and I. Nwogu, “Word-conditioned 3d american sign language motion generation,” in *Findings of the Association for Computational Linguistics: EMNLP*, 2024.
- [13] S. Stoll, N. C. Camgöz, S. Hadfield, and R. Bowden, “Text2sign: Towards sign language production using neural machine translation and generative adversarial networks,” *Int. J. Comput. Vis.*, 2020.
- [14] S. Fang, C. Sui, Y. Zhou, X. Zhang, H. Zhong, Y. Tian, and C. Chen, “Signdiff: Diffusion model for american sign language production,” in *19th IEEE International Conference on Automatic Face and Gesture Recognition, FG*, 2025.
- [15] F. Qi, Y. Duan, H. Zhang, and C. Xu, “Signgen: End-to-end sign language video generation with latent diffusion,” in *European Conference on Computer Vision*, 2024.
- [16] S. Fang, Y. Feng, H. Zhong, Y. Zhang, and D. N. Metaxas, “Stable signer: Hierarchical sign language generative model,” in *Annual Meeting of the Association for Computational Linguistics*, 2026.
- [17] J. Han, J. Liu, Y. Jiang, B. Yan, Y. Zhang, Z. Yuan, B. Peng, and X. Liu, “Infinity: Scaling bitwise autoregressive modeling for high-resolution image synthesis,” in *Conference on Computer Vision and Pattern Recognition*, 2025.
- [18] J. Liu, J. Han, B. Yan, H. Wu, F. Zhu, X. Wang, Y. Jiang, B. Peng, and Z. Yuan, “Infinitystar: Unified spacetime autoregressive modeling for visual generation,” in *Conference and Workshop on Neural Information Processing Systems*, 2025.
- [19] C. Chang, F. Sanches, G. Gao, and M. Liarokapis, “On human to robot skill transfer for the execution of complex tactile american sign language tasks with a bimanual robot platform,” in *46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBC*, 2024.
- [20] X. Tan, N. Bai, H. Gardener, Z. Zhong, L. Zhang, L. Yang, Z. Duan, M. Galeitsiwe, and Z. Tang, “Signvla: A gloss-free vision-language-action framework for real-time sign language-guided robotic manipulation,” *arXiv*, 2026.
- [21] G. Qiao, S. Lin, R. Zuo, Z. Wu, K. Jia, and G. Liu, “Signbot: Learning human-to-humanoid sign language interaction,” in *International Conference on Robotics and Automation*, 2026.
- [22] A. van den Oord, O. Vinyals, and K. Kavukcuoglu, “Neural discrete representation learning,” in *Conference and Workshop on Neural Information Processing Systems*, 2017.
- [23] L. Xue, N. Constant, A. Roberts, M. Kale, R. Al-Rfou, A. Siddhant, A. Barua, and C. Raffel, “mt5: A massively multilingual pre-trained text-to-text transformer,” in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2021, Online, June 6-11, 2021*, 2021.
- [24] A. C. Duarte, S. Palaskar, L. Ventura, D. Ghadiyaram, K. DeHaan, F. Metze, J. Torres, and X. Giró-i-Nieto, “How2sign: A large-scale multimodal dataset for continuous american sign language,” in *Conference on Computer Vision and Pattern Recognition*, 2021.
- [25] W. Yin, Z. Cai, R. Wang, F. Wang, C. Wei, H. Mei, W. Xiao, Z. Yang, Q. Sun, A. Yamashita, Z. Liu, and L. Yang, “WHAC: world-grounded humans and cameras,” in *European Conference on Computer Vision*, 2024.
- [26] J. Romero, D. Tzionas, and M. J. Black, “Embodied hands: modeling and capturing hands and bodies together,” *ACM Trans. Graph.*, 2017.
- [27] R. A. Potamias, J. Zhang, J. Deng, and S. Zafeiriou, “Wilor: End-to-end 3d hand localization and reconstruction in-the-wild,” in *Conference on Computer Vision and Pattern Recognition*, 2025.
- [28] I. G. S. Team, “Granite 4.1 speech,” 2026. [Online]. Available: <https://huggingface.co/ibm-granite/granite-speech-4.1-2b>
- [29] H. Sakoe and S. Chiba, “Dynamic programming algorithm optimization for spoken word recognition,” *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 1978.
- [30] K. Papineni, S. Roukos, T. Ward, and W. Zhu, “Bleu: a method for automatic evaluation of machine translation,” in *Annual Meeting of the Association for Computational Linguistics*, 2002.